

# Exact-Polar Muon vs. Finite-Step Newton–Schulz Muon

## Abstract

This note compares two versions of Muon’s orthogonalization step. The usual implementation in this repository uses a finite number of quintic Newton–Schulz steps, which approximates the polar factor. The exact-polar ablation computes an SVD and sets every live singular value to one exactly. The experiment asks whether Muon’s behavior comes from the exact polar ideal  $UV^\top$ , or from the softer finite-step Newton–Schulz approximation.

## 1 The Two Orthogonalizers

Let  $M \in \mathbb{R}^{m \times n}$  be the momentum or Nesterov momentum matrix. Write its compact singular value decomposition as

$$M = U\Sigma V^\top, \quad \Sigma = \text{diag}(\sigma_1, \dots, \sigma_k), \quad k = \min(m, n).$$

**Exact-polar Muon.** The exact polar update is

$$D_{\text{svd}} = UV^\top.$$

For every live singular direction, this sets

$$\sigma_i(D_{\text{svd}}) = 1.$$

This is the literal “put all singular values to one” version.

**Newton–Schulz Muon.** The practical Muon baseline avoids the SVD. It first normalizes

$$X_0 = \frac{M}{\|M\|_F + \varepsilon}$$

and then applies the quintic Newton–Schulz-style iteration

$$X_{j+1} = 2X_j - 1.5(X_j X_j^\top)X_j + 0.5(X_j X_j^\top)^2 X_j.$$

On singular values, this is the scalar map

$$q(\sigma) = 2\sigma - 1.5\sigma^3 + 0.5\sigma^5.$$

The repository’s standard Muon config uses 12 such steps.

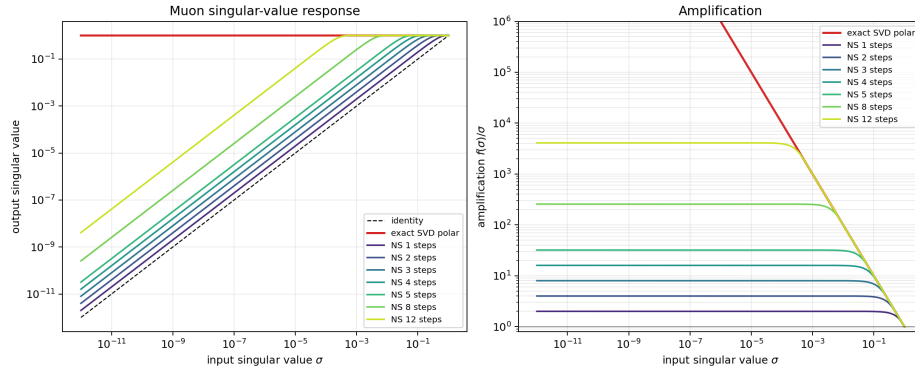


Figure 1: Singular-value response of exact-polar Muon versus finite-step quintic Newton–Schulz Muon. The exact SVD polar update sends every live singular value to 1. The NS approximation reaches 1 quickly for moderate singular values, but leaves very small directions below 1 after a finite number of steps.

| input $\sigma$ | NS, 5 steps         | NS, 12 steps        | exact SVD polar |
|----------------|---------------------|---------------------|-----------------|
| $10^{-8}$      | $3.2 \cdot 10^{-7}$ | $4.1 \cdot 10^{-5}$ | 1               |
| $10^{-7}$      | $3.2 \cdot 10^{-6}$ | $4.1 \cdot 10^{-4}$ | 1               |
| $10^{-6}$      | $3.2 \cdot 10^{-5}$ | $4.1 \cdot 10^{-3}$ | 1               |
| $10^{-5}$      | $3.2 \cdot 10^{-4}$ | $4.1 \cdot 10^{-2}$ | 1               |
| $10^{-4}$      | $3.2 \cdot 10^{-3}$ | $3.9 \cdot 10^{-1}$ | 1               |
| $10^{-3}$      | $3.2 \cdot 10^{-2}$ | 1.0                 | 1               |

Table 1: Representative singular-value outputs for the repository’s Muon Newton–Schulz polynomial versus exact SVD polar orthogonalization.

## 2 Scalar Response

The exact-polar path maps every nonzero singular value directly to 1. In contrast, finite-step Newton–Schulz only approaches this behavior. For very small singular values, the scalar map is approximately linear:

$$q(\sigma) \approx 2\sigma \quad \text{when } \sigma \ll 1.$$

After 12 steps, tiny singular values are therefore amplified by about

$$2^{12} = 4096,$$

not sent all the way to 1.

### 3 Experiment

The code now exposes this as a Muon option:

`orthogonalize`  $\in$  `{ns,svd}`.

The baseline uses

`orthogonalize: ns,      ns_steps: 12.`

The exact-polar ablation uses

`orthogonalize: svd.`

Both variants keep the same momentum, Nesterov update, shape-aware learning rate scaling, weight decay, and AdamW fallback groups. Thus the intended comparison is finite-step NS soft polar versus exact SVD polar, where the exact SVD path sets all live singular values to one.